An Experimental Framework for Life Expectancy Prediction from Single-Country Life-Factor Data

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Abstract. Life-expectancy prediction from person-level risk factors is usually studied on a small cohort from one country, which limits data volume and transferability. We propose an experimental framework that combines any person-wise life-factor dataset with public life tables to predict life expectancy for the reference country and for every country with a life table. A Kyrgyz cohort of 95 participants, collected with a cyber-physical system, is labelled with the remaining life expectancy $e(x)$ of the 2023 Kyrgyz life table. It is modelled with a mean baseline, a direct life-table lookup, Ridge, Random Forest (RF) and Gradient Boosting (GB), and re-labelled with the tables of 192 countries. A built-in audit separates circular target construction from validation design. Because the labels are a function of age, the lookup reproduces them exactly, and the cross-validated R^2 of 0.96 of the learned models comes from age (97–98% of SHAP). Without age and its proxies, RF and GB do not beat the mean baseline, a limitation of the labelling scheme rather than evidence against lifestyle effects. A separate life-table forecaster, evaluated at five rolling origins (2019–2023), found naive persistence hardest to beat (pooled MAE 0.79 years), and feature ablations trace the learned models' failures to momentum features.

Keywords: Life expectancy · Data leakage · SHAP · Life tables · Transferability · Kyrgyzstan

1 Introduction

Life expectancy is studied at two levels. At the *population* level, demographic and Bayesian models project it for every country [7,18,20], and machine-learning (ML) models on country indicators now report R^2 above 0.98 [2,5,6]. At the *individual* level, lifestyle risk factors such as smoking, diet, physical activity and body mass index can separate survival by decades within one country [10]. Wearables and smart scales now make such person-level data cheaper to collect [3].

The two levels rarely meet. Individual-level models assume one domestic mortality baseline, while cross-country studies find that structural and country-level factors, not personal habits, explain most of the variation in life expectancy [6, 21, 22]. A model built on one country's cohort therefore faces three problems: person-level data are *scarce*, results do not *generalise* automatically to other

mortality regimes, and they are hard to *reproduce* and audit. Age is a legitimate predictor of remaining life expectancy, but when the target *is* the life-table value for the participant's age, a model that sees age recovers the table, and its accuracy reflects this circularity [11] rather than a learned effect of lifestyle.

We generalise a Kyrgyz study in which a cyber-physical system with a smart scale and lifestyle monitoring collected risk factors [3] into a proposed experimental framework that links person-level data to population life tables. Its contributions are: (i) a pipeline that anchors any person-wise life-factor dataset to a reference life table and re-labels it under any country's mortality schedule; (ii) an audit that separates circular target construction from validation design; (iii) rolling-origin forecasting of life tables for 192 countries against simple baselines [9], with feature ablations; (iv) per-country status and feature importance (Appendix A).

2 Related Work

Population-level forecasting. Classical methods extrapolate age-specific mortality, most prominently Lee–Carter [12]. The UN adopted Bayesian hierarchical projections of life expectancy for all countries [18]. The Global Burden of Disease study forecasts and estimates mortality for 195–204 countries [7, 20]. Multi-population models show that stable relationships often hold only within clusters of similar countries [13], and trajectories diverge through shocks such as COVID-19 [1]. For short horizons, the naive forecast remains the standard benchmark [9].

ML on country indicators. Tree ensembles [4, 8] and stacking models trained on country panels reach high accuracy with drivers such as income, health expenditure, undernourishment and governance [2, 5, 21, 24]. The determinants differ by development level: communicable disease and basic living conditions matter most in low-income countries, and non-communicable disease and behaviour in high-income ones [22,24]. Attribution analyses find that country-level structure dominates, which argues for country-held-out validation before claims of generalisation [6].

Individual level and validation. Life-table models extended with lifestyle factors quantify personal risk within one country [10], and sensor-based systems collect such factors continuously [3]. Leakage inflates reported performance when target information enters the features or the validation split [11]. Grouped cross-validation is recommended for clustered data [19], and permutation tests [16] and SHAP [14, 15] check whether and how a model uses its inputs. Our framework bridges these lines and builds such checks into the pipeline.

3 Proposed Experimental Framework

Figure 1 shows the framework. Its inputs are *any* person-wise life-factor table with age and risk factors, and a set of abridged life tables; nothing is specific to Kyrgyzstan or to the chosen features. Its two components share only the life

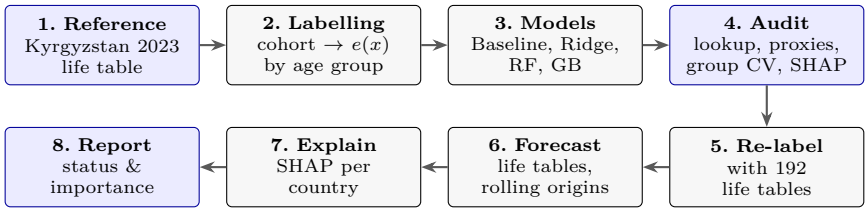


Fig. 1. Proposed experimental framework. Person-wise life-factor data from one country are labelled with that country's life table, modelled and audited, then re-labelled and forecast for every country with a life table.

tables. The *cohort component* (steps 1–5) labels and models person-level data. The *life-table component* (step 6) forecasts the tables from their own history and uses no cohort data. It supplies tables for years not yet published, which step 2 can use to label a cohort for a future year. All steps use scikit-learn [17] with fixed seeds.

Design goals. The framework *addresses*, but does not remove, three problems. (1) *Data scarcity*: life tables label the cohort for 192 countries without new data, but a small cohort limits which effects can be learned (Sect. 4.2). (2) *Generalisability*: re-labelling the same participants only rescales their labels; showing that person-level relations hold elsewhere requires cohorts from other countries. (3) *Reproducibility*: public life tables, fixed seeds and one script regenerate all results, given access to the cohort data.

Data. The cohort [3] has 95 participants and 30 variables. They are aged 16–65 (median 19; 63 aged 15–19), 24% are female, and mean BMI is 22.2 ± 4.0. We use the study's 12 features: age, sex, BMI, sleep, steps per day, physical activity, smoking, alcohol, body water, muscle mass, body fat and chronic disease. Cleaning fixes the decimal format and trailing spaces that created spurious categories, encodes physical activity ordinally and imputes one missing muscle-mass value with the median. The abridged life tables come from the United Nations World Population Prospects 2024 [23]. They cover 192 countries for 2015–2023 with 19 age groups, giving q_x, l_x and e_x (no missing values), and are used as published; preprocessing only parses age groups and builds lagged panels. Code: `https://github.com/Data-Analysis-and-Research-Institute/life-expectancy-kg-to-192`. Ethics: non-invasive, anonymised data from volunteers; no formal approval was sought [**TODO: exemption rule; consent, incl. parental for ages 16–17**].

Labelling and models. For country c and year t, participant i of age a_i receives $y_i^{(c)} = e_{x(a_i)}^{(c,t)}$ with $x(a) = \max\{x_k : x_k \le a\}$. The reference is Kyrgyzstan in 2023, which replaces the fixed table of [3]; that table lies about 2.8 years above the 2023 values at young ages. Two reference predictors bracket the learned models: a mean baseline, and a direct *life-table lookup* $\hat{y}_i = e_{x(a_i)}$, which reproduces the labels exactly. Between them we train Ridge, RF (200 trees) and GB (200 stages, learning rate 0.05; library defaults otherwise), evaluated

on an 80/20 split, as in [3], and with 5×5 repeated k-fold CV (mean of the 25 fold MAEs). Augmentation appends two copies of the 76 training rows with Gaussian noise (SD 1% of each feature's training SD) on the seven continuous features (age, BMI, sleep, steps, body water, muscle mass, body fat); labels are unchanged and the 19 test rows stay clean.

Target and validation audit. We separate two issues. (i) *Circular target construction*: y is a step function of age, so a model with access to age or its proxies can recover the lookup. This is not train–test contamination. Augmentation is train-only, and no fitted statistic crosses folds except the median imputation of one value. (ii) *Validation design*: random folds place participants of the same age group, who share one label, in both training and test sets, so models interpolate between known labels. Leave-one-age-group-out CV instead tests prediction at unseen ages. Regularised RF and GB (depth ≤ 4, ≥ 5 samples per leaf) are run in five scenarios: (A) all features, random 5-fold CV; (B) all features, age-group CV; (C) no age; (D) no age and no age proxies; (E) D with age-group CV. Proxies have $|\rho_{\text{Spearman}}| \geq 0.2$ with age, a threshold in the largest gap of the correlation profile (BMI 0.35, body fat 0.24, all others ≤ 0.14); we also test 0.1, 0.3 and 0.4. One 5-fold split and pooled out-of-fold predictions are used throughout. Gains over the mean baseline get 95% intervals from a paired bootstrap over participants (2000 resamples). Permutation tests refit each model on 100 permuted targets with the same folds, with $p = (1 + \#\{\text{MAE}_{\text{perm}} \leq \text{MAE}_{\text{obs}}\})/101$ (minimum 0.01). Random-CV scenarios shuffle labels across participants; age-group-CV scenarios permute which label each group carries, because a global shuffle would break the constant-label groups and give an easier null.

Re-labelling and forecasting. The cohort is re-labelled with each country's 2023 table, a counterfactual that asks what this cohort's labels would be under another mortality schedule. For life-table forecasting, e_x in year t is predicted from $e_{t-1..t-3}$, their changes Δ, $\log q_{x,t-1}$, the country's e^0_{t-1} and its change, and the age group. Learned models predict $e_t - e_{t-1}$. We compare naive persistence, a per-series linear trend, and global Ridge, RF and histogram GB at every feasible origin (2019–2023; 2018 is the first year with three lags), each trained only on earlier years. A *selected* forecaster picks the model with the lowest mean error at earlier origins, and ablations remove one feature group at a time. Four series with a single-year e^0 change above 10 years (Central African Republic, Holy See, South Sudan, State of Palestine; next largest 6.4) are excluded and reported separately.

4 Results

Starting point. Kyrgyz e^0 fell 1.7 years in 2020 and reached 71.8 in 2023, rank 116 of 192 and 1.9 years below the world median; Kazakhstan, Russia, Uzbekistan and Tajikistan are ahead and Turkmenistan behind.

Table 1. Cohort models, Kyrgyzstan 2023 target (years): 80/20 hold-out, augmented hold-out, 5×5 repeated CV, and MAE reported in [3].

Model	MAE	RMSE	R^2	MAE (aug.)	CV MAE	CV R^2	MAE [3]
Life-table lookup	0.00	0.00	1.00	0.00	0.00 ± 0.00	1.00	–
Baseline (mean)	7.28	11.08	0.00	7.28	6.24 ± 1.35	−0.11	–
Ridge	1.24	1.59	0.98	1.19	1.21 ± 0.23	0.96	0.95
Random Forest	0.82	2.71	0.94	0.51	0.66 ± 0.54	0.95	0.39
Gradient Boosting	**0.56**	2.00	0.97	**0.45**	**0.52** ± 0.53	0.96	0.18

4.1 Cohort Models

Table 1 reproduces the ranking of [3]: GB, then RF, then Ridge, all far better than the mean baseline. The direct lookup, however, is exact (MAE 0, R^2 1), so the learned models only approximate a function known in closed form. Our errors exceed those reported, consistent with the corrected life table and the clean test set. With *age alone*, GB reaches a CV MAE of 0.20 years; without age, the MAE rises to 4.4–4.8 years.

4.2 Target and Validation Audit

Table 2 uses regularised models and one pooled split, so its errors differ from Table 1, which uses unregularised models and averages 25 fold MAEs (scenario A: RF 1.87, GB 0.99 vs. 0.66, 0.52). With age (A), RF and GB recover most of the lookup, and age takes 97–98% of the mean |SHAP|. At unseen ages (B), GB still beats the group-permutation null ($p = 0.01$) and RF does not ($p = 0.17$). Without age (C), a gain remains, traced by SHAP to the proxy BMI (3.2 years in RF, 6.1 in GB). Without proxies (D, E), neither model beats the baseline, and all intervals include zero.

Pooled age-group errors are dominated by the 15–19 group (63 of 95; Table 3); when it is held out, only 32 older participants remain for training. Macro-averaged over groups, GB errs by 4.6 years in B against 17.4 for the baseline, while in E all models lie within 0.1 years of it. The result is robust to the threshold: removing only BMI (0.3) leaves gains of 1–7% with intervals spanning zero, 0.1 leaves both models at or below the baseline, and only keeping all proxies (0.4) restores a gain (22–36%). These negative results are a property of the labelling scheme, not evidence against lifestyle effects: participants in an age group share one label, so lifestyle can predict it only through its correlation with age.

4.3 Re-labelling with 192 Life Tables

Under other countries' schedules, the cohort's mean label ranges from 43.5 to 68.8 years (Kyrgyzstan: 53.5). In all 192 label sets, age takes about 98% of GB's SHAP importance and the model without age and proxies never beats the baseline (Appendix A). With participants fixed, this shows the audit holds under any mortality schedule, *not* that the models generalise to other populations.

Table 2. Target and validation audit: pooled out-of-fold MAE (years), gain over the mean baseline (95% bootstrap interval) and permutation p.

Scenario	Base	RF	GB	RF gain % [CI]	GB gain % [CI]	p RF/GB
A. All, random	6.25	1.87	0.99	70 [63, 78]	84 [79, 90]	–
B. All, age-group	12.31	6.48	4.70	47 [43, 52]	62 [57, 66]	0.17/0.01
C. No age, random	6.25	4.61	4.09	26 [13, 38]	35 [19, 48]	0.01/0.01
D. No age/prox., random	6.25	6.40	7.02	−2 [−11, 5]	−12 [−29, 1]	0.27/0.46
E. No age/prox., age-group	12.31	12.49	12.66	−1 [−5, 2]	−3 [−12, 6]	0.56/0.54

Table 3. Leave-one-age-group-out CV: group size, label and MAE (years) per held-out age group for scenarios B (all features) and E (no age, no proxies). The baseline is the same in B and E.

Age group	15–19	20–24	25–29	30–34	35–39	40–44	45–49	50–54	65–69	Macro
n	63	17	1	4	2	2	3	1	2	
$e(x)$	58.16	53.30	48.48	43.66	38.91	34.34	29.96	25.72	14.21	
Baseline	13.79	0.26	5.09	10.29	14.91	19.59	24.32	28.09	40.15	17.39
RF (B)	6.85	4.10	3.84	4.90	3.26	3.03	6.66	10.05	24.37	7.45
GB (B)	4.69	4.43	0.78	4.05	3.89	1.29	3.85	0.63	17.98	4.62
RF (E)	13.58	2.24	3.89	10.37	12.56	20.44	24.66	29.57	38.65	17.33
GB (E)	13.43	3.68	5.55	10.91	9.11	21.77	26.97	29.56	36.08	17.45

4.4 Life-Table Forecasting

On the 188 stable series (Table 4), learned models are marginally better only in the calm year 2019; persistence or Ridge wins 2020–21, the trend the 2022 rebound, and persistence 2023. Pooled, persistence has the lowest MAE, RMSE and 95th-percentile error and is best in 107 countries, including Kyrgyzstan (MAE 0.82). Selection on earlier origins switches between trend and persistence and averages 1.09 over 2020–2023, against 0.94 for always using persistence.

MAE versus RMSE. On all 192 countries, persistence had the lowest 2023 MAE (0.69) but GB the lowest RMSE (1.66 vs. 1.85). The gap comes from the four unstable series: persistence carries the Central African Republic's 2022 value (18.9 years) into 2023 (published 57.4), a 38-year error that dominates RMSE. Without them, persistence also has the lowest 2023 RMSE (0.80 vs. 1.04), so no trade-off remains; on them, no model is reliable (MAE 2–7 years at every origin). The Central African Republic's published e^0 for 2015–2023 reads 51.9, 51.0, 45.1, 52.2, 31.5, 50.6, 40.1, 18.9, 57.4, passed unchanged by our preprocessing. The earlier RF projection of 35.3 years for 2024 was the momentum features' reversal of the +38.5-year jump, so we report no projections for unstable series.

Ablations. SHAP ranks a recent change (Δ_{t-2}) first in 107 countries; ablations show that this reliance hurts. Removing the momentum features lowers GB's mean MAE from 0.92 to 0.89, and a level-only model (age group, e_{t-1}) to 0.84 (RF: 0.93 to 0.87), mostly in 2021–23. Removing the country's e^0 raises it (0.96); deeper lags and $\log q$ have no effect. The models fail by extrapolating recent changes across regime shifts, and even the level-only model does not beat persistence (0.84 vs. 0.79).

Table 4. Rolling-origin forecasting of e_x, 188 stable series × 19 ages (years). Wins: countries with the lowest mean MAE. Selected: best model at earlier origins.

	MAE by forecast year					Pooled 2019–23			
Model	2019	2020	2021	2022	2023	MAE	RMSE	P95	Wins
Naive (persistence)	0.19	**0.85**	0.88	1.46	**0.56**	**0.79**	**1.21**	**2.68**	107
Linear trend	**0.17**	0.95	1.11	**1.09**	1.06	0.88	1.27	2.70	29
Ridge	0.18	0.93	**0.83**	1.77	0.76	0.89	1.35	3.05	16
Random Forest	**0.17**	0.93	0.97	1.82	0.78	0.93	1.43	3.23	18
Gradient Boosting	**0.17**	0.93	0.94	1.77	0.79	0.92	1.40	3.21	18
Selected (earlier years)	–	0.95	0.88	1.46	1.06				

5 Discussion and Conclusion

The framework produces reproducible life-table labels for one cohort under 192 mortality schedules. Its audit shows that a life-table target measures how well a model reads the table: the lookup is exact, and the accuracy of such designs, including [3], comes from age. The weak lifestyle-only models reflect this labelling scheme, not an absence of the lifestyle effects reported elsewhere [6, 10, 22]. Feature effects need a target that varies between individuals, such as survival or health outcomes collected by the cyber-physical system, and population claims need cohorts from other countries. For the life tables, persistence was the strongest forecaster across origins, so the latest published table is the best basis for next-year labels, and ML forecasters must be reported against it. *Limitations*: a small, young, urban cohort; tables not split by sex; only five forecast origins, four in or after COVID-19.

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A Prediction Status and Feature Importance for Every Country

Kyrgyzstan comes first, then countries by e^0_{2023}. *Status:* ΔKGZ is the gap to Kyrgyzstan; $\geq$2019 means e^0_{2023} is at or above its 2019 level; Best/MAE give the forecaster with the lowest mean MAE (years) over the 2019–2023 rolling origins; Naive is the persistence MAE; Band is a *retrospective error band* of that MAE (High < 0.5, Med. < 1.5, Low), not an independently validated measure of forecastability; † marks unstable source series, which are not evaluated. *Importance* (mean |SHAP| of GB): Age% is age's share in the full cohort model; Non-age top is the strongest other feature; LF gain is the CV MAE gain of the lifestyle-only model (no age, BMI, body fat) over the mean baseline, where negative means no skill; LF top-2 are its largest features; Fcst. top is the top forecasting feature (share %). Full values are in `outputs/tables/`.

Table 5: Prediction status and feature importance for all 192 countries (columns explained above).

Country	e^0_{23}	Rank	ΔKGZ	$\geq$2019	Best	MAE	Naive	Band	Age%	Non-age top	LF gain	LF top-2	Fcst. top (%)
Kyrgyzstan	71.8	116	+0.0	yes	Naive	0.82	0.82	Med.	98.5	body fat	-12	sleep, water	Δ(t-1) (32)
Holy See†	88.7	1	+17.0	yes	–	–	–	Unstable	98.2	body fat	-12	water, sleep	Δ(t-2) (75)
Rep. of Korea	85.1	2	+13.4	yes	Trend	0.65	0.80	Med.	98.4	body fat	-13	sleep, water	e0(t-1) (57)
Japan	85.1	3	+13.3	yes	Ridge	0.29	0.41	High	98.5	activity	-13	water, sleep	e0(t-1) (55)
Andorra	84.8	4	+13.0	yes	Naive	1.51	1.51	Low	98.5	activity	-12	sleep, water	Δ(t-2) (35)
Singapore	84.3	5	+12.5	yes	Naive	0.28	0.28	High	98.5	muscle	-13	water, sleep	e0(t-1) (29)
Switzerland	84.2	6	+12.4	yes	Trend	0.44	0.58	High	98.4	activity	-12	sleep, water	Δ(t-2) (53)
Australia	84.1	7	+12.3	yes	Trend	0.52	0.68	Med.	98.6	body fat	-12	water, sleep	e0(t-1) (35)
Malta	84.0	8	+12.3	yes	Naive	0.53	0.53	Med.	98.5	activity	-13	water, sleep	e0(t-1) (42)
Spain	83.9	9	+12.1	yes	Trend	0.63	0.83	Med.	98.5	activity	-12	water, sleep	Δ(t-2) (59)
Italy	83.9	10	+12.1	yes	Trend	0.71	0.92	Med.	98.4	body fat	-13	water, sleep	Δ(t-2) (43)
France	83.5	11	+11.8	yes	Trend	0.40	0.41	High	98.3	activity	-12	water, sleep	e0(t-1) (43)
Sweden	83.4	12	+11.7	yes	Trend	0.35	0.47	High	98.3	activity	-12	sleep, water	Δ(t-2) (48)
UAE	83.4	13	+11.6	yes	Ridge	1.35	1.54	Med.	98.4	body fat	-11	water, sleep	Δ(t-2) (68)
Norway	83.2	14	+11.5	yes	Ridge	0.24	0.33	High	98.4	activity	-12	water, sleep	e0(t-1) (42)
Canada	82.9	15	+11.1	yes	Trend	0.46	0.59	High	98.4	body fat	-13	water, sleep	e0(t-1) (31)
Qatar	82.8	16	+11.0	no	Naive	0.96	0.96	Med.	98.1	body fat	-11	water, sleep	Δ(t-2) (38)
Iceland	82.7	17	+10.9	no	RF	0.33	0.61	High	98.6	activity	-12	water, sleep	Δe0(t-1) (43)
Ireland	82.7	18	+10.9	no	Trend	0.39	0.44	High	98.4	activity	-13	water, sleep	Δ(t-2) (39)
Portugal	82.6	19	+10.8	yes	Naive	0.45	0.45	High	98.4	activity	-12	water, sleep	e0(t-1) (43)
Israel	82.6	20	+10.8	no	Naive	0.19	0.19	High	98.4	body fat	-13	water, sleep	Δ(t-2) (36)
Luxembourg	82.4	21	+10.6	yes	Naive	0.32	0.32	High	98.4	activity	-12	sleep, water	Δe0(t-1) (41)
Belgium	82.3	22	+10.5	yes	Trend	0.59	0.81	Med.	98.6	muscle	-13	water, sleep	Δ(t-2) (57)
Netherlands	82.2	23	+10.5	yes	Naive	0.36	0.36	High	98.4	body fat	-13	water, sleep	Δ(t-2) (40)
New Zealand	82.2	24	+10.4	yes	Ridge	0.50	0.72	Med.	98.4	activity	-13	water, sleep	Δe0(t-1) (54)
Denmark	82.0	25	+10.3	yes	Trend	0.26	0.28	High	98.4	activity	-12	water, sleep	Δ(t-1) (31)
Finland	82.0	26	+10.2	no	Ridge	0.26	0.34	High	98.5	body fat	-13	water, sleep	Δe0(t-1) (32)
Greece	82.0	27	+10.2	yes	Ridge	0.45	0.74	High	98.4	muscle	-13	water, sleep	Δ(t-2) (51)
Austria	81.9	28	+10.1	no	Trend	0.34	0.43	High	98.4	muscle	-13	water, sleep	Δe0(t-1) (36)
Slovenia	81.8	29	+10.1	yes	Naive	0.51	0.51	Med.	98.5	activity	-13	water, sleep	Δ(t-2) (40)
Cyprus	81.7	30	+10.0	yes	Ridge	0.31	0.47	High	98.4	muscle	-13	water, sleep	Δ(t-2) (36)
Bahrain	81.6	31	+9.9	yes	Naive	1.16	1.16	Med.	98.4	body fat	-12	water, sleep	Δ(t-2) (39)
Germany	81.6	32	+9.8	yes	Trend	0.37	0.54	High	98.4	muscle	-12	water, sleep	e0(t-1) (33)
Chile	81.5	33	+9.7	yes	Naive	0.85	0.85	Med.	98.5	activity	-13	water, sleep	Δ(t-2) (40)
United Kingdom	81.4	34	+9.7	no	Trend	0.48	0.52	High	98.4	muscle	-11	water, sleep	Δ(t-2) (37)
Costa Rica	81.2	35	+9.4	yes	RF	0.83	0.95	Med.	98.5	muscle	-12	water, sleep	Δ(t-2) (51)
Maldives	81.2	36	+9.4	yes	Naive	0.93	0.93	Med.	98.4	body fat	-11	sleep, water	Δe0(t-1) (30)
Kuwait	80.6	37	+8.8	yes	Ridge	1.46	1.47	Med.	98.5	water	-12	water, sleep	Δ(t-2) (58)

(continued)

Country	e_{23}^0	Rank	ΔKGZ	≥2019	Best	MAE	Naive	Band	Age%	Non-age top	LF gain	LF top-2	Fcst. top (%)
Panama	80.1	38	+8.3	yes	Naive	1.14	1.14	Med.	98.4	body fat	-11	water, sleep	e0(t-1) (31)
Oman	80.0	39	+8.3	yes	Naive	1.42	1.42	Med.	98.3	body fat	-13	sleep, water	Δ(t-2) (39)
Anguilla	79.9	40	+8.2	yes	Naive	0.29	0.29	High	98.5	water	-12	water, sleep	e0(t-1) (22)
Czechia	79.9	41	+8.1	yes	Naive	0.89	0.89	Med.	98.4	activity	-12	water, sleep	Δ(t-2) (34)
Albania	79.7	42	+8.0	yes	Naive	1.03	1.03	Med.	98.5	water	-13	water, sleep	Δ(t-1) (36)
USA	79.5	43	+7.7	yes	Naive	0.91	0.91	Med.	98.5	body fat	-12	water, sleep	e0(t-1) (31)
Estonia	79.3	44	+7.5	yes	Trend	0.52	0.66	Med.	98.4	activity	-12	water, sleep	Δ(t-2) (46)
Saudi Arabia	79.0	45	+7.2	yes	GB	0.58	0.65	Med.	98.4	activity	-13	water, sleep	Δ(t-2) (34)
Poland	78.8	46	+7.0	yes	Naive	0.99	0.99	Med.	98.3	body fat	-12	sleep, water	Δ(t-2) (33)
Croatia	78.6	47	+6.8	yes	Naive	0.47	0.47	High	98.4	activity	-12	water, sleep	Δ(t-1) (39)
Slovakia	78.5	48	+6.7	yes	GB	1.26	1.39	Med.	98.5	activity	-12	sleep, water	Δ(t-2) (60)
Uruguay	78.3	49	+6.5	yes	GB	0.84	1.22	Med.	98.4	muscle	-12	water, sleep	Δ(t-2) (70)
Cuba	78.2	50	+6.5	yes	Naive	1.65	1.65	Low	98.5	activity	-12	sleep, water	Δ(t-2) (67)
Peru	78.2	51	+6.5	yes	GB	1.90	2.10	Low	98.3	body fat	-12	water, sleep	Δ(t-2) (62)
Jordan	78.1	52	+6.3	yes	Naive	1.22	1.22	Med.	98.5	body fat	-12	water, sleep	Δ(t-2) (46)
China	78.0	53	+6.3	no	Naive	0.12	0.12	High	98.4	activity	-12	water, sleep	Δ(t-2) (40)
Bosnia & Herz.	78.0	54	+6.2	yes	Naive	1.35	1.35	Med.	98.4	activity	-12	water, sleep	Δ(t-2) (51)
Lebanon	77.9	55	+6.1	no	Naive	1.50	1.50	Low	98.4	body fat	-12	water, sleep	Δ(t-2) (59)
Antigua & Barbuda	77.8	56	+6.1	yes	Naive	0.14	0.14	High	98.6	body fat	-12	water, sleep	Δe0(t-1) (37)
Colombia	77.8	57	+6.1	yes	Naive	1.48	1.48	Med.	98.5	body fat	-12	sleep, water	Δ(t-2) (52)
Iran	77.8	58	+6.1	yes	Naive	1.26	1.26	Med.	98.3	activity	-12	sleep, water	e0(t-1) (41)
Sri Lanka	77.6	59	+5.8	yes	Trend	0.34	0.40	High	98.4	activity	-12	water, sleep	Δ(t-2) (46)
Ecuador	77.6	60	+5.8	no	Naive	1.86	1.86	Low	98.3	water	-13	water, sleep	Δ(t-1) (32)
Argentina	77.5	61	+5.7	yes	Ridge	1.02	1.07	Med.	98.5	body fat	-12	sleep, water	Δ(t-2) (50)
North Macedonia	77.5	62	+5.7	yes	Naive	1.28	1.28	Med.	98.4	activity	-12	water, sleep	Δ(t-2) (41)
Montenegro	77.2	63	+5.5	yes	Ridge	1.11	1.16	Med.	98.7	body fat	-12	sleep, water	Δ(t-2) (58)
Türkiye	77.2	64	+5.5	no	Naive	0.87	0.87	Med.	98.5	water	-12	water, sleep	Δ(t-1) (39)
Hungary	77.1	65	+5.3	yes	Ridge	0.93	0.94	Med.	98.5	body fat	-12	sleep, water	Δ(t-2) (51)
Malaysia	77.0	66	+5.2	yes	Ridge	0.79	1.02	Med.	98.4	body fat	-12	water, sleep	Δ(t-2) (52)
Serbia	76.9	67	+5.1	yes	Naive	1.30	1.30	Med.	98.5	activity	-13	sleep, water	Δ(t-2) (46)
Thailand	76.7	68	+5.0	no	Ridge	0.43	0.55	High	98.2	body fat	-12	sleep, water	e0(t-1) (56)
Tunisia	76.6	69	+4.8	yes	Naive	1.16	1.16	Med.	98.5	activity	-13	water, sleep	Δ(t-2) (56)
Aruba	76.6	70	+4.8	yes	Naive	0.88	0.88	Med.	98.6	muscle	-13	water, sleep	Δ(t-2) (54)
Algeria	76.4	71	+4.6	yes	Naive	1.00	1.00	Med.	98.3	activity	-13	water, sleep	Δ(t-2) (40)
Latvia	76.3	72	+4.5	yes	RF	0.92	1.06	Med.	98.5	body fat	-12	sleep, water	Δ(t-2) (58)
Cabo Verde	76.2	73	+4.4	yes	Naive	0.73	0.73	Med.	98.4	body fat	-12	water, sleep	Δe0(t-1) (55)
Barbados	76.1	74	+4.4	no	Ridge	0.18	0.32	High	98.5	activity	-12	water, sleep	e0(t-1) (46)
Lithuania	76.1	75	+4.3	no	Ridge	0.73	0.78	Med.	98.6	BMI	-12	sleep, water	e0(t-1) (47)
Romania	76.0	76	+4.2	yes	Naive	1.04	1.04	Med.	98.6	body fat	-13	water, sleep	Δ(t-2) (37)
Brazil	75.9	77	+4.2	no	Naive	0.94	0.94	Med.	98.4	body fat	-11	water, sleep	Δ(t-2) (34)
Armenia	75.8	78	+4.1	yes	Naive	1.59	1.59	Low	98.6	body fat	-12	sleep, water	e0(t-1) (52)
Bulgaria	75.6	79	+3.9	yes	Ridge	1.28	1.35	Med.	98.6	body fat	-13	sleep, water	Δ(t-2) (47)
Brunei	75.5	80	+3.7	yes	GB	0.53	0.86	Med.	98.5	body fat	-11	water, sleep	Δe0(t-1) (59)
Morocco	75.4	81	+3.7	yes	Naive	0.66	0.66	Med.	98.5	activity	-12	water, sleep	Δe0(t-1) (35)
Grenada	75.3	82	+3.5	yes	Naive	0.29	0.29	High	98.5	body fat	-13	sleep, water	Δ(t-2) (41)
Mexico	75.2	83	+3.5	yes	Naive	1.75	1.75	Low	98.3	water	-13	water, sleep	Δe0(t-1) (27)
Mauritius	75.1	84	+3.3	yes	GB	0.51	0.91	Med.	98.4	water	-12	sleep, water	Δ(t-2) (71)
Nicaragua	75.1	85	+3.3	yes	Naive	1.44	1.44	Med.	98.5	body fat	-12	sleep, water	Δe0(t-1) (31)
Bangladesh	75.0	86	+3.2	yes	Naive	1.07	1.07	Med.	98.1	muscle	-11	sleep, water	Δe0(t-1) (31)
Bahamas	74.7	87	+3.0	yes	Naive	1.37	1.37	Med.	98.3	steps	-12	sleep, water	Δ(t-2) (37)
Viet Nam	74.7	88	+2.9	yes	Trend	0.54	0.61	Med.	98.5	body fat	-12	water, sleep	Δ(t-2) (70)
Georgia	74.6	89	+2.8	yes	RF	0.91	0.98	Med.	98.5	water	-12	sleep, water	Δ(t-2) (45)
Kazakhstan	74.5	90	+2.7	yes	Naive	1.47	1.47	Med.	98.2	water	-12	sleep, water	Δ(t-2) (37)
Azerbaijan	74.5	91	+2.7	yes	Naive	1.24	1.24	Med.	98.3	activity	-11	water, sleep	Δe0(t-1) (28)
Belarus	74.4	92	+2.7	yes	Naive	0.78	0.78	Med.	98.4	body fat	-13	sleep, water	Δe0(t-1) (28)
Paraguay	73.9	93	+2.1	yes	GB	1.80	1.93	Low	98.4	steps	-12	sleep, water	Δ(t-2) (67)
Dominican Republic	73.8	94	+2.0	yes	Naive	0.85	0.85	Med.	98.4	activity	-12	water, sleep	Δe0(t-1) (30)
Suriname	73.8	95	+2.0	yes	Trend	1.46	1.71	Med.	98.4	water	-13	water, sleep	Δ(t-2) (65)
DPR Korea	73.7	96	+1.9	yes	Naive	0.19	0.19	High	98.4	body fat	-12	water, sleep	Δ(t-2) (37)
Ukraine	73.7	97	+1.9	no	RF	0.93	1.17	Med.	98.3	body fat	-13	sleep, water	Δ(t-2) (64)
Belize	73.6	98	+1.9	yes	Naive	0.72	0.72	Med.	98.6	water	-12	sleep, water	Δ(t-2) (27)

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Country	e_{23}^0	Rank	ΔKGZ	≥2019	Best	MAE	Naive	Band	Age%	Non-age top	LF gain	LF top-2	Fcst. top (%)
Trinidad & Tobago	73.6	99	+1.8	yes	Naive	0.70	0.70	Med.	98.4	muscle	-12	water, sleep	Δ(t-2) (43)
Russia	73.3	100	+1.6	yes	Naive	1.34	1.34	Med.	98.4	body fat	-13	water, sleep	Δ(t-2) (36)
Honduras	73.0	101	+1.2	yes	Naive	1.06	1.06	Med.	98.4	body fat	-12	sleep, water	Δ(t-2) (32)
Bhutan	73.0	102	+1.2	yes	Trend	0.05	0.14	High	98.5	steps	-12	water, sleep	e0(t-1) (56)
Seychelles	72.9	103	+1.1	yes	Trend	1.31	1.35	Med.	98.5	water	-11	water, sleep	Δ(t-2) (87)
Tonga	72.7	104	+1.0	yes	Naive	0.19	0.19	High	98.6	body fat	-12	water, sleep	e0(t-1) (32)
Guatemala	72.7	105	+0.9	yes	Naive	1.48	1.48	Med.	98.4	activity	-11	water, sleep	Δ(t-2) (58)
Saint Lucia	72.6	106	+0.9	yes	Naive	1.24	1.24	Med.	98.5	activity	-13	sleep, water	Δ(t-2) (56)
Venezuela	72.6	107	+0.8	no	Naive	0.42	0.42	High	98.4	activity	-12	water, sleep	e0(t-1) (36)
Iraq	72.4	108	+0.6	yes	GB	0.66	0.79	Med.	98.6	body fat	-12	water, sleep	e0(t-1) (44)
Uzbekistan	72.4	109	+0.6	yes	Trend	0.47	0.49	High	98.5	activity	-13	water, sleep	Δ(t-1) (29)
Syria	72.3	110	+0.5	yes	Naive	0.60	0.60	Med.	98.5	body fat	-12	water, sleep	Δ(t-1) (30)
St Kitts & Nevis	72.2	111	+0.5	yes	RF	0.50	0.87	High	98.6	activity	-13	water, sleep	Δ(t-2) (66)
El Salvador	72.1	112	+0.4	yes	Naive	0.72	0.72	Med.	98.3	activity	-13	sleep, water	Δ(t-1) (42)
India	72.1	113	+0.4	yes	Naive	1.49	1.49	Med.	98.7	body fat	-12	water, sleep	Δ(t-2) (60)
Tajikistan	71.9	114	+0.1	yes	RF	0.82	0.98	Med.	98.4	activity	-12	water, sleep	Δe0(t-1) (56)
Mongolia	71.8	115	+0.0	yes	Naive	0.83	0.83	Med.	98.6	body fat	-12	sleep, water	Δ(t-2) (49)
Egypt	71.7	117	-0.1	yes	Naive	0.94	0.94	Med.	98.3	activity	-13	sleep, water	Δ(t-2) (40)
Samoa	71.7	118	-0.1	yes	Naive	0.22	0.22	High	98.5	body fat	-12	sleep, water	Δ(t-2) (43)
Jamaica	71.5	119	-0.3	no	Naive	0.86	0.86	Med.	98.4	body fat	-12	water, sleep	Δ(t-2) (62)
Dominica	71.4	120	-0.4	no	Trend	0.45	0.45	High	98.5	body fat	-12	water, sleep	Δ(t-2) (45)
Vanuatu	71.4	121	-0.4	yes	Naive	0.41	0.41	High	98.5	body fat	-12	water, sleep	Δ(t-2) (35)
St Vincent & Gren.	71.4	122	-0.4	yes	Naive	0.63	0.63	Med.	98.6	activity	-12	water, sleep	Δe0(t-1) (29)
Indonesia	71.2	123	-0.6	yes	Naive	1.18	1.18	Med.	98.7	body fat	-13	water, sleep	Δ(t-2) (47)
Moldova	71.2	124	-0.6	yes	Naive	0.84	0.84	Med.	98.4	body fat	-13	sleep, water	Δ(t-2) (40)
Cambodia	70.8	125	-1.0	yes	Naive	0.40	0.40	High	98.7	body fat	-12	water, sleep	Δ(t-2) (36)
Solomon Islands	70.5	126	-1.2	yes	Naive	0.47	0.47	High	98.4	body fat	-12	sleep, water	Δ(t-2) (26)
Palau	70.4	127	-1.4	yes	Naive	0.23	0.23	High	98.3	body fat	-12	sleep, water	Δ(t-2) (48)
Nepal	70.3	128	-1.4	yes	RF	0.52	0.52	Med.	98.3	body fat	-12	sleep, water	Δ(t-2) (30)
Guyana	70.3	129	-1.5	yes	Naive	1.84	1.84	Low	98.8	body fat	-12	water, sleep	Δ(t-2) (64)
Turkmenistan	70.2	130	-1.6	yes	Naive	0.33	0.33	High	98.3	body fat	-12	water, sleep	Δ(t-2) (41)
Philippines	69.9	131	-1.9	yes	Trend	1.14	1.24	Med.	98.5	body fat	-13	sleep, water	Δ(t-2) (64)
Sao Tome & Pr.	69.8	132	-2.0	yes	Naive	0.38	0.38	High	98.6	body fat	-13	sleep, water	Δe0(t-1) (27)
Libya	69.4	133	-2.3	no	GB	0.83	1.00	Med.	98.4	water	-12	sleep, water	Δe0(t-1) (33)
Yemen	69.3	134	-2.4	yes	RF	0.58	0.68	Med.	98.3	activity	-11	water, sleep	Δe0(t-1) (28)
Botswana	69.2	135	-2.5	yes	Trend	1.76	1.83	Low	98.6	body fat	-12	sleep, water	Δ(t-2) (63)
Lao PDR	69.0	136	-2.8	yes	Trend	0.26	0.30	High	98.6	body fat	-13	water, sleep	Δ(t-2) (35)
Tuvalu	68.9	137	-2.8	yes	Naive	0.65	0.65	Med.	98.6	body fat	-12	water, sleep	Δe0(t-1) (27)
Senegal	68.7	138	-3.1	yes	GB	0.40	0.46	High	98.4	water	-12	water, sleep	Δ(t-2) (44)
Eritrea	68.7	139	-3.1	yes	GB	0.38	0.38	High	98.4	body fat	-11	sleep, water	Δ(t-2) (35)
Bolivia	68.6	140	-3.1	yes	Naive	2.32	2.32	Low	98.5	water	-12	water, sleep	Δ(t-2) (45)
Mauritania	68.5	141	-3.3	yes	Naive	0.49	0.49	High	98.6	body fat	-13	sleep, water	Δe0(t-1) (41)
Gabon	68.4	142	-3.4	yes	Naive	0.32	0.32	High	98.5	body fat	-13	sleep, water	Δ(t-2) (43)
Uganda	68.3	143	-3.4	yes	Naive	0.40	0.40	High	98.4	body fat	-12	sleep, water	Δe0(t-1) (26)
Rwanda	67.8	144	-3.9	yes	Trend	0.12	0.22	High	98.5	body fat	-13	sleep, water	Δ(t-2) (48)
Timor-Leste	67.7	145	-4.1	yes	Trend	0.30	0.37	High	98.6	activity	-13	water, sleep	Δ(t-2) (42)
Pakistan	67.7	146	-4.1	yes	Naive	0.54	0.54	Med.	98.6	activity	-12	water, sleep	Δe0(t-1) (42)
Namibia	67.5	147	-4.2	yes	RF	1.35	1.73	Med.	98.3	body fat	-12	water, sleep	Δ(t-2) (63)
Ethiopia	67.4	148	-4.4	yes	Naive	0.46	0.46	High	98.6	steps	-12	water, sleep	e0(t-1) (33)
Malawi	67.4	149	-4.4	yes	RF	0.52	0.55	Med.	98.7	body fat	-12	sleep, water	e0(t-1) (38)
Fiji	67.3	150	-4.4	yes	Naive	0.75	0.75	Med.	98.7	BMI	-13	sleep, water	Δ(t-2) (51)
Tanzania	67.1	151	-4.7	yes	Trend	0.30	0.40	High	98.3	body fat	-12	water, sleep	Δ(t-2) (43)
Myanmar	66.9	152	-4.9	yes	GB	0.30	0.39	High	98.2	activity	-11	water, sleep	Δ(t-2) (29)
Comoros	66.8	153	-4.9	yes	Naive	0.38	0.38	High	98.5	body fat	-12	sleep, water	e0(t-1) (29)
Kiribati	66.5	154	-5.2	yes	Naive	0.75	0.75	Med.	98.6	body fat	-12	water, sleep	Δ(t-2) (35)
Zambia	66.4	155	-5.4	yes	RF	0.80	0.82	Med.	98.4	muscle	-11	water, sleep	Δ(t-2) (31)
Papua New Guinea	66.2	156	-5.6	yes	GB	0.47	0.48	High	98.6	activity	-12	water, sleep	e0(t-1) (35)
South Africa	66.2	157	-5.6	no	Naive	1.59	1.59	Low	98.4	body fat	-13	sleep, water	Δ(t-2) (55)
Sudan	66.1	158	-5.6	yes	Naive	0.59	0.59	Med.	98.4	activity	-11	water, sleep	e0(t-1) (32)
Afghanistan	66.1	159	-5.7	yes	Naive	1.22	1.22	Med.	98.5	body fat	-12	water, sleep	e0(t-1) (32)
Djibouti	66.0	160	-5.7	yes	Naive	0.49	0.49	High	98.5	body fat	-12	water, sleep	e0(t-1) (49)

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Country	e_{23}^0	Rank	ΔKGZ	≥2019	Best	MAE	Naive	Band	Age%	Non-age top	LF gain	LF top-2	Fcst. top (%)
Gambia	65.9	161	-5.8	yes	GB	0.31	0.48	High	98.4	body fat	-11	water, sleep	e0(t-1) (39)
Congo	65.8	162	-5.9	yes	Naive	0.45	0.45	High	98.5	body fat	-13	water, sleep	e0(t-1) (42)
Ghana	65.5	163	-6.2	yes	Naive	0.28	0.28	High	98.5	activity	-11	water, sleep	e0(t-1) (57)
Palestine†	65.2	164	-6.5	no	–	–	–	Unstable	98.5	activity	-10	water, sleep	Δ(t-2) (42)
Haiti	65.0	165	-6.8	yes	GB	0.52	0.63	Med.	98.5	body fat	-12	water, sleep	Δ(t-2) (25)
Angola	64.7	166	-7.1	yes	Naive	0.34	0.34	High	98.7	body fat	-13	sleep, water	e0(t-1) (39)
Eswatini	64.2	167	-7.6	yes	GB	1.26	1.36	Med.	98.5	body fat	-12	water, sleep	Δ(t-2) (47)
Guinea-Bissau	64.1	168	-7.6	yes	Naive	0.65	0.65	Med.	98.3	body fat	-12	sleep, water	Δe0(t-1) (47)
Equatorial Guinea	63.8	169	-8.0	yes	Naive	0.31	0.31	High	98.3	body fat	-12	sleep, water	e0(t-1) (42)
Cameroon	63.7	170	-8.0	yes	RF	0.49	0.56	High	98.3	body fat	-13	sleep, water	Δe0(t-1) (37)
Burundi	63.7	171	-8.0	yes	RF	0.32	0.34	High	98.4	body fat	-11	water, sleep	Δ(t-2) (35)
Madagascar	63.7	172	-8.1	yes	GB	0.35	0.40	High	98.3	activity	-11	water, sleep	e0(t-1) (36)
Kenya	63.6	173	-8.1	yes	Naive	0.63	0.63	Med.	98.3	body fat	-12	water, sleep	Δe0(t-1) (33)
Mozambique	63.6	174	-8.1	yes	Naive	0.78	0.78	Med.	98.4	water	-12	sleep, water	Δ(t-2) (37)
Nauru	62.9	175	-8.9	no	Naive	0.39	0.39	High	98.3	water	-12	sleep, water	Δ(t-2) (49)
Togo	62.8	176	-9.0	yes	Naive	0.26	0.26	High	98.3	activity	-12	water, sleep	e0(t-1) (47)
Zimbabwe	62.8	177	-9.0	yes	Naive	0.67	0.67	Med.	98.5	body fat	-12	sleep, water	Δ(t-2) (41)
Liberia	62.2	178	-9.6	yes	Naive	0.21	0.21	High	98.5	activity	-12	water, sleep	e0(t-1) (50)
Côte d'Ivoire	62.0	179	-9.8	yes	Naive	0.39	0.39	High	98.4	BMI	-12	water, sleep	e0(t-1) (52)
DR Congo	61.9	180	-9.8	yes	RF	0.28	0.40	High	98.3	activity	-12	water, sleep	e0(t-1) (35)
Sierra Leone	61.9	181	-9.9	yes	Naive	0.31	0.31	High	98.3	body fat	-13	sleep, water	e0(t-1) (50)
Niger	61.2	182	-10.6	yes	RF	0.28	0.39	High	98.4	activity	-12	water, sleep	e0(t-1) (45)
Burkina Faso	61.1	183	-10.6	yes	RF	0.22	0.28	High	98.5	water	-12	water, sleep	e0(t-1) (34)
Benin	60.8	184	-11.0	yes	Naive	0.28	0.28	High	98.4	body fat	-11	water, sleep	e0(t-1) (36)
Guinea	60.8	185	-11.0	yes	Naive	0.30	0.30	High	98.3	muscle	-11	water, sleep	e0(t-1) (51)
Mali	60.4	186	-11.3	yes	Naive	0.33	0.33	High	98.4	activity	-11	sleep, water	e0(t-1) (54)
Somalia	58.9	187	-12.9	yes	GB	0.94	1.14	Med.	98.3	body fat	-12	sleep, water	Δ(t-2) (45)
South Sudan†	58.1	188	-13.7	no	–	–	–	Unstable	98.4	activity	-12	water, sleep	e0(t-1) (40)
Lesotho	57.4	189	-14.4	yes	Naive	0.81	0.81	Med.	98.5	body fat	-12	sleep, water	Δ(t-2) (44)
Central African Rep.†	57.4	190	-14.4	yes	–	–	–	Unstable	98.3	body fat	-12	water, sleep	Δ(t-2) (70)
Chad	55.4	191	-16.4	yes	RF	0.41	0.41	High	98.4	activity	-11	water, sleep	e0(t-1) (60)
Nigeria	54.5	192	-17.3	yes	Trend	0.21	0.23	High	98.3	activity	-12	sleep, water	e0(t-1) (45)